

Пробная олимпиада — 3

Time allowed: 3 hours

1. A linear operator $A: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is given by $A(x) = [c, x]$, where c is a fixed vector and $[\cdot, \cdot]$ denotes the cross product. Find all eigenvalues of A .
2. Find all real numbers α, β for which the limit

$$\lim_{x, y \rightarrow 0+} \frac{x^{2\alpha} y^{2\beta}}{x^{2\alpha} + y^{3\beta}}$$

exists and is finite.

3. For every $0 < \alpha < 1$ let $R(\alpha) \subset \mathbb{R}^2$ be the region bounded by the convex pentagon with vertices $(0, 1 - \alpha)$, $(\alpha, 0)$, $(1, 0)$, $(1, 1)$ and $(0, 1)$. Let R be the set of points belonging to every $R(\alpha)$ with $0 < \alpha < 1$, that is, $R = \bigcap_{0 < \alpha < 1} R(\alpha)$. Determine the area of R .
4. An n -dimensional cube is given. Consider all the segments connecting any two different vertices of the cube. How many distinct intersection points do these segments have (the vertices themselves being excluded)?
5. Call a nonzero vector in $\mathbb{R}^3$ with integer coordinates a *lattice vector*. Prove that every lattice vector of length at most 2021 is perpendicular to some lattice vector of length at most 100.